

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated: 12th March, 2026
Semester: 1st	Session: 2026
Lab/Project/Assignment #: 5	CLOs to be covered: CLO2
Lab Title: Functions	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO2	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Mark s	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 5: Functions

Lab Goals/Objectives:

To learn how to define and call user-defined functions in Python, including functions with parameters, return values, default arguments, and variable-length arguments.

Equipment Required:

Python 3, VS Code / PyCharm

Theory:

A function is a reusable block of code that performs a specific task. Functions are defined using the def keyword, may accept parameters (including default and variable-length *args parameters), and may return a value using the return statement.

Lab Work:**Task 1: Larger Number and its Multiplication Table**

```
num1 = int(input("Enter value of 1st number: "))
```

```
num2 = int(input("Enter value of 2nd number: "))
```

```
def compare(num1, num2):
```

```
    return max(num1, num2)
```

```
lambda max: max(num1, num2)
```

```
    max = max(num1, num2)
```

```
    print("The greater number is", max)
```

```
for i in range(1, 15):
```

```
    product = max * i
```

```
    print(product)
```

```
    i+=1
```

Result:

```
Enter value of 1st number: 13
Enter value of 2nd number: 15
The greater number is 15
15
30
45
60
75
90
105
120
135
150
165
180
195
210
```

Task 2: Fahrenheit and Celsius Conversions

```
temp = float(input("Enter temperature here="))  
unit = input("Enter units here( 'C' for Celsius and 'F' for Fahrenheit)=")
```

```
def celsius_to_fahrenheit(c):  
    return ((c*9/5)+32)
```

```
def fahrenheit_to_celsius(f):  
    return ((f-32)*5/9)
```

```
if unit.upper()=="C":  
    result = fahrenheit_to_celsius(temp)  
    print("Temperature in Celsius is ",result)  
elif unit.upper() == "F":  
    result = celsius_to_fahrenheit(temp)  
    print("Temperature in Fahrenheit is ", result)  
else:  
    print("Invalid unit. Enter F or C")
```

Result:

```
Enter temperature here=34  
Enter units here( 'C' for Celsius and 'F' for Fahrenheit)=f  
Temperature in Fahrenheit is 93.2
```

Task 3: UDF Greet Function

```
def greet(hello):  
    return hello("Welcome to Python Programming")  
status= str(input("Please enter your learning status: 'Beginner' or 'Experienced'"))  
if status == "Beginner" or status == "beginner":  
    greet(print)  
elif status == "Experienced" or status == "experienced":  
    greet(print)  
else:  
    greet(print)
```

Result:

```
Please enter your learning status: 'Beginner' or 'Experienced'b  
Welcome to Python Programming
```

Task 4: Name Function

```
def hello(name):  
    return ("Hello," + name)  
name = str(input("Please enter your name: "))  
print(hello(name))
```

Result:

```
Please enter your name: ann  
Hello,ann
```

Task 5: Add a and b

```
def add(a,b):  
    return a+b  
a=float(input("Please enter your first number: "))  
b=float(input("Please enter your second number: "))  
print(add(a,b))
```

Result:

```
Please enter your first number: 23  
Please enter your second number: 2  
25.0  
Please enter your number: 2
```

Task 6: Exponent and Default Exponent

```
base = float(input("Enter your base number: "))  
exp = (input("Enter your exponent or press enter for default exponent: "))
```

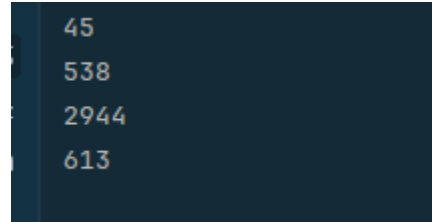
```
def power(base,exp):  
    return base**exp  
  
if exp == "":  
    exp=2  
else:  
    exp = int(exp)  
print(power(base,exp))
```

Result:

```
Please enter your number: 2  
4  
Enter your base number: 4  
Enter your exponent or press enter for default exponent:  
16.0  
Please enter your number: 2
```

Task 7: Total(*numbers)

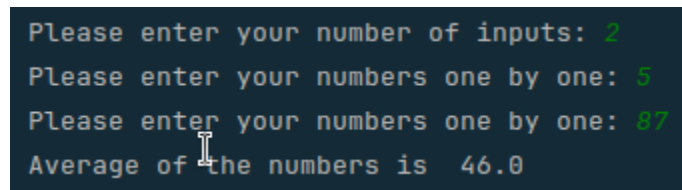
```
def sum_all(*args):  
    return sum(args)  
print (sum_all(1,2,3,4,5,6,7,8,9))  
print (sum_all(193,345))  
print(sum_all(112,366,864,345,234,456,567))  
print (sum_all(26,36,67,38,95,85,54,65,63,32,52))
```

Result:

```
45  
538  
2944  
613
```

Task 8: Average(*numbers)

```
sum=0  
inputs= int(input("Please enter your number of inputs: "))  
for i in range (1,inputs+1):  
    num= float(input("Please enter your numbers one by one: "))  
    sum+=num  
def avg(sum,inputs):  
    return sum/inputs  
print("Average of the numbers is ",(avg(sum,inputs)))
```

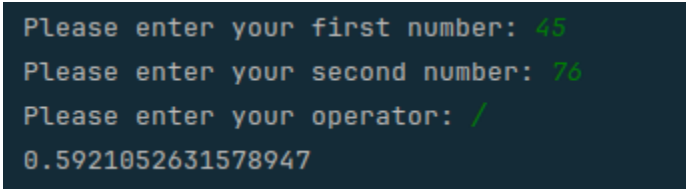
Result:

```
Please enter your number of inputs: 2  
Please enter your numbers one by one: 5  
Please enter your numbers one by one: 87  
Average of the numbers is 46.0
```

Task 9: Calculator

```
def add(a,b):  
    return a+b  
def sub(a,b):  
    return a-b  
def mul(a,b):  
    return a*b  
def div(a,b):  
    return a/b  
a = float(input("Please enter your first number: "))  
b = float(input("Please enter your second number: "))  
operator = str(input("Please enter your operator: "))
```

```
if operator == "+":
    print(add(a,b))
elif operator == "-":
    print(sub(a,b))
elif operator == "*":
    print(mul(a,b))
elif operator == "/":
    print(div(a,b))
else:
    print("Operator is not correct")
```

Result:

```
Please enter your first number: 45
Please enter your second number: 76
Please enter your operator: /
0.5921052631578947
```

Conclusions:

Multiple user-defined functions were created and called with different types of arguments, deepening the understanding of function definition, parameters, and return values.